

The empirical recurrence for OEIS A235073: recovering a conjecture that is not written down

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Abstract

OEIS A235073 records “Empirical recurrence of order 73”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 392$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 73, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 73 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235073 is “Number of $(n+1) \times (2+1)$ 0..7 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 3, with no adjacent elements equal (constant-stress tilted 1×1 tilings)”. It has offset 1 and begins

1828, 12228, 86704, 644188, 4832976, 36598068, 278216484, 2124989472, ...

The entry states:

Empirical recurrence of order 73 (see link above).

As of the “Last modified” line on the live entry (Jun 19 2022 23:23:45, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 392$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 392$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 784$ exact terms of the model returns order 73 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{73} c_i a(n-i),$$

c_1	9	c_{20}	−2283632847474963	c_{39}	−3329463683883615540161	c_{58}	−225744944363
c_2	253	c_{21}	15605178960194553	c_{40}	245300716516507608409	c_{59}	−27782235168
c_3	−2296	c_{22}	18458905148435130	c_{41}	5481194002809115052659	c_{60}	609939038385
c_4	−28151	c_{23}	−122955462773333188	c_{42}	−963122871609885453735	c_{61}	10497031775
c_5	255768	c_{24}	−122446251676533541	c_{43}	−7522195683906434951171	c_{62}	−12447468817
c_6	1859886	c_{25}	806629225402213328	c_{44}	2058458454461970952639	c_{63}	80990559867
c_7	−16727287	c_{26}	667182131877191467	c_{45}	8549214399966799766011	c_{64}	18684407986
c_8	−83009548	c_{27}	−4430940721105867228	c_{46}	−3155362747161882170284	c_{65}	−2436756607
c_9	729387541	c_{28}	−2978297375229876665	c_{47}	−7979548285655389768333	c_{66}	−1980874602
c_{10}	2685919408	c_{29}	20462103965419263597	c_{48}	3706593864463403134970	c_{67}	3638060206
c_{11}	−22793267893	c_{30}	10806687134235708830	c_{49}	6051487249587236761280	c_{68}	1387484385
c_{12}	−65779647686	c_{31}	−79643524358625973344	c_{50}	−3413453559203876872794	c_{69}	−317114374
c_{13}	534767403988	c_{32}	−31319886445936922252	c_{51}	−3677271288867846726924	c_{70}	−566931184
c_{14}	1254116051910	c_{33}	261603403683159192127	c_{52}	2482175229170949527036	c_{71}	152226383
c_{15}	−9727483891431	c_{34}	69680882992309107234	c_{53}	1756318505495499297736	c_{72}	99756319
c_{16}	−18975638746037	c_{35}	−725171880256128246186	c_{54}	−1424899389632322832824	c_{73}	−30694252
c_{17}	140416341968064	c_{36}	−106357306795746363554	c_{55}	−640058920803422146240		
c_{18}	230976261260999	c_{37}	1694461108573024804395	c_{56}	642618504247472447952		
c_{19}	−1636592446458350	c_{38}	56813379667179363992	c_{57}	168478657601714799584		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 73, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $73 + 4$ terms removed returns order 73 as well, so $\deg q_{\min} = 73 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $73 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 73 that this sequence satisfies — and that family turns out to have exactly one member.

References

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